

# ZATCA Fatoora Phase 2 E-Invoicing Compliance Architecture

SECTION 04-compliance-ksa    TYPE policy    STATUS **active**    OWNER Compliance Lead / CTO  
UPDATED 2026-08-30

This document establishes the technical, operational, and regulatory architecture for **ZATCA (Zakat, Tax and Customs Authority) Fatoora Phase 2 Integration** across BDA Company International's platforms (Fly GACA and Captain Adel).

## 1. Corporate & Tax Entity Identity

All invoices generated by Fly GACA B2B billing and consumer subscriptions strictly inherit from corporate facts in `01-governance/company-facts.md` :

| Parameter                     | Certified Value                                                      |
|-------------------------------|----------------------------------------------------------------------|
| Legal Name (Arabic)           | شركة بدع الدولية                                                     |
| Legal Name (English)          | BDA Company International                                            |
| Commercial Registration (CR)  | 7030976893                                                           |
| VAT Registration Number       | 311415259500003                                                      |
| Taxpayer Identification (TIN) | 3114152595                                                           |
| Registered National Address   | 2816 King Fahd Rd, Al Sahafah Dist., Riyadh 13321-6548, Saudi Arabia |
| Standard Tax Rate             | 15% Standard VAT                                                     |

## 2. Invoicing Classification & Flows

```
graph TD
  A[Billing Trigger: B2B Cohort / Stripe / Moyasar] --> B{Invoice Category}
  B -->|B2B Flight Academy / Enterprise| C[Standard Tax Invoice - 0100000]
  B -->|B2C Pilot Subscriptions / Prep Packs| D[Simplified Tax Invoice - 0200000]
  C --> E[Generate UBL 2.1 XML]
  D --> E
  E --> F[Canonical SHA-256 Digest Hashing]
  F --> G[ECDSA Cryptographic Signing with CSID]
  G --> H[Tag-Length-Value TLV QR Code Encoding]
  H --> I[ZATCA Fatoora Clearance / Reporting API]
  I --> J[Archived with Cryptographic Chaining PIH]
```

## 2.1 Standard Tax Invoices (B2B - Code 388 / 01000000 )

- **Use Case:** Flight Academy bulk seat licenses (e.g., OxfordSaudia, SNAA, Saudi Aramco Aviation).
- **Clearance Workflow:** XML is transmitted to ZATCA Fatoora Clearance API before delivery to the customer. Once approved, the signed XML with clearance status is delivered.
- **Buyer Identifiers:** Mandatory Buyer Name, Buyer National Address, and Buyer VAT Number (where registered).

## 2.2 Simplified Tax Invoices (B2C - Code 388 / 02000000 )

- **Use Case:** Individual student pilots purchasing Pro annual passes or AIP Exam Prep bundles.
- **Reporting Workflow:** Generated immediately with Phase 2 TLV QR code and reported asynchronously to ZATCA within 24 hours.

---

## 3. Cryptographic & Data Structures

### 3.1 Tag-Length-Value (TLV) QR Code (Phase 2 Specification)

The QR code embedded on all generated PDF and HTML invoices encodes a 9-tag Base64 binary payload:

1. **Tag 1:** Seller Name ( BDA Company International )
2. **Tag 2:** Seller VAT Number ( 311415259500003 )
3. **Tag 3:** Invoice Timestamp ( YYYY-MM-DDTHH:MM:SSZ )
4. **Tag 4:** Invoice Total with VAT ( SAR X.XX )
5. **Tag 5:** Total VAT Amount ( SAR X.XX )
6. **Tag 6:** SHA-256 Invoice XML Digest
7. **Tag 7:** ECDSA Digital Signature (using secp256k1 / NIST P-256)
8. **Tag 8:** Seller Public Key (from CSID)
9. **Tag 9:** ZATCA Certificate Authority Signature

### 3.2 Cryptographic Chaining (Previous Invoice Hash - PIH)

Every invoice header includes a `<cac:AdditionalDocumentReference>` element containing the SHA-256 hash of the immediate preceding invoice ( PIH ). The genesis invoice uses a certified fixed seed hash

( NWZLY2ViNjZmZmM4NmYzOGQ5NTI3ODZjNmQ2OTZjNzljMmRiYzIzOWRkNGU5MWI0NjAzZTQ4MmUwNzMzNGZhNw== ), establishing an immutable audit ledger compliant with anti-tampering mandates.

---

## 4. Implementation Reference

---

The production implementation resides in:

- Engine: `FlyGACA/server/src/zatca-core.ts`
  - Automated Verification: `FlyGACA/server/tests/zatca-core.test.ts` (4 unit tests verifying TLV encoding, SHA-256 digest, UBL 2.1 generation, and Academy B2B billing).
  - Tax Invoice Template: `Office/03-finance/tax-invoice-template.html`
- 

## 5. Record Retention & Archival

---

Per ZATCA Article 66, all XML invoices, cryptographic logs, and verification responses must be archived in encrypted, in-Kingdom storage (Google Cloud Platform `me-central2` Dammam region) for a statutory period of **not less than 6 years (72 months)**.